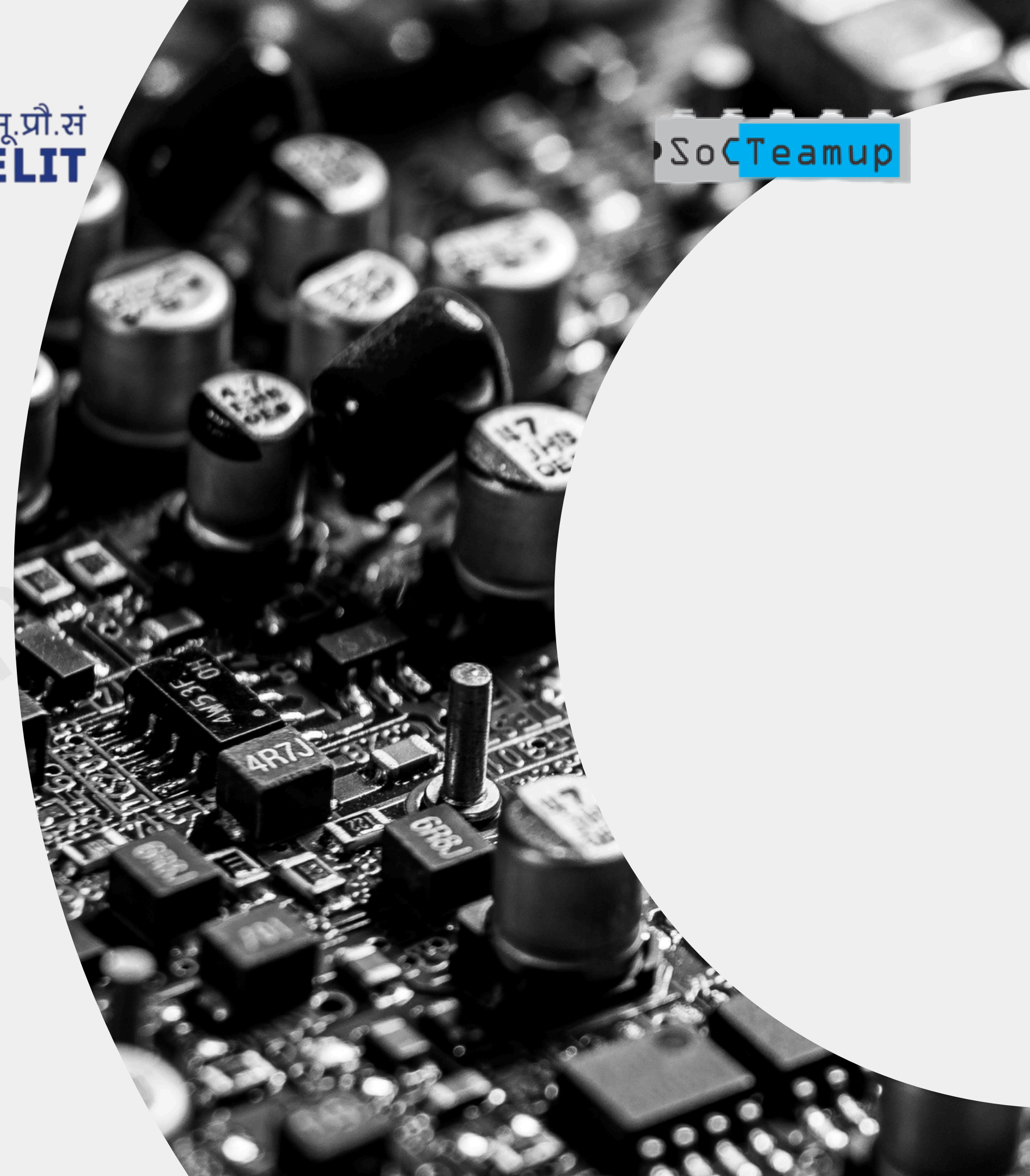


Role of timing constraints (SDC) and Technology libraries (.lib files)





Outline

Importance of timing constraints in synthesis
What is the SDC format?
Why is SDC written in Tcl?
Basic Tcl commands used in SDC
Clock definition in SDC
Input/output delay constraints
Special paths (false, multicycle)
SDC for synthesis vs STA
What is a .lib file?
Cell delays, drive strength, timing arcs
.lib and technology-specific modeling
Delay calculation using .lib
How tools use SDC + .lib together
Writing realistic timing constraints
Summary and Best Practices

Importance of timing constraints in synthesis

Synthesis needs to know:

- What timing goals to meet
- Clock frequency and I/O timing
- Which paths to ignore

Timing constraints guide:

- Logic optimization
- Gate sizing
- Buffer insertion

Result: Meet setup/hold constraints across the design

What is the SDC Format?

SDC = Synopsys Design Constraints

- SDC is the standard TCL-based file format for specifying timing, power, and design rules in digital flows.
- Key Purpose
- Defines clocks, input/output delays, false paths, and multicycle paths for synthesis, STA, and PnR tools.
- Ensures design meets performance targets across the RTL-to-GDSII flow.

Main Commands

`create_clock`: Sets clock period, waveform on ports/nets.

`set_input_delay/set_output_delay`: External timing relative to reference clock.

`set_max_delay/set_multicycle_path`: Path-specific timing exceptions.

`set_false_path`: Ignores non-functional paths during analysis.

➔ Why SDC Uses Tcl Syntax

Tcl = Tool Command Language

- Widely used in EDA tools
- Easy to read and extend
- Built-in support in synthesis and STA tools

Example:

create_clock, set_input_delay, set_false_path → all
are Tcl commands

- Tools: Yosys, Vivado, Design Compiler all support Tcl

Key Tcl Constructs in SDC

```
create_clock -period 10 [get_ports clk]
set_input_delay 2 -clock clk [get_ports din]
set_false_path -from [get_ports debug]
```

- Commands use Tcl's syntax: command options [arguments]
- get_ports, get_cells, get_clocks are Tcl queries
- SDC is just a specialized Tcl script

➔ Defining Clock Constraints in SDC

```
create_clock -period 10 [get_ports clk]
```

- Tells the tool that clock has a period of 10ns
- Required for all setup/hold checks
- Can also define generated clocks, virtual clocks

Input and Output Delay Constraints

```
set_input_delay 2 -clock clk [get_ports a]  
set_output_delay 3 -clock clk [get_ports y]
```

- Models delay from/to external components
- Important for interface timing (e.g., sensors, memory)

False Paths and Multicycle Paths

```
set_false_path -from [get_ports debug_en]  
set_multicycle_path 2 -from U1/ff1 -to U2/ff2
```

- False Path: not functionally timed
- Multicycle Path: valid every N cycles
- Avoid over-constraining paths

SDC for **Synthesis** vs STA

SDC files guide both synthesis optimization and STA verification, but with different priorities and detail levels

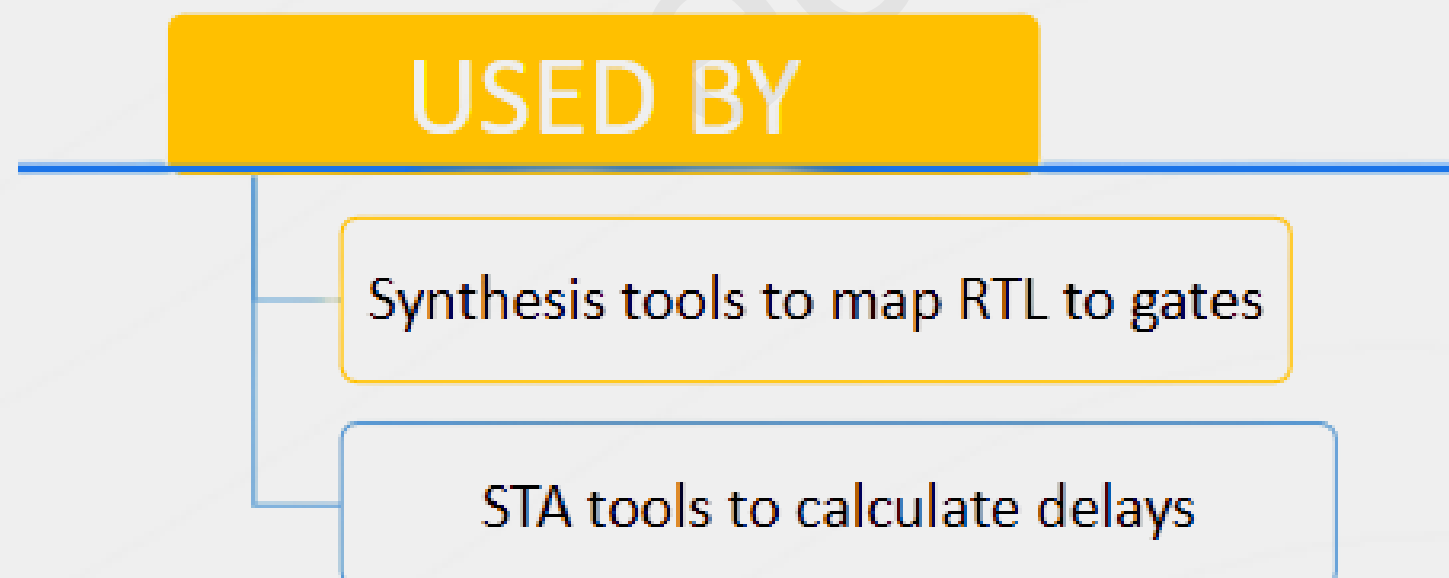
- **Synthesis SDC:** Guides optimization with basic clocks/I/O delays; uses wireload estimates for fast logic mapping and retiming.
- **STA SDC:** Verifies signoff timing with complete exceptions, precise virtual clocks, and full path coverage.
- **Key Difference:** Synthesis prioritizes speed (simplified), STA demands accuracy (comprehensive) for tapeout validation.

What is a .lib File?

.lib = Liberty format cell library

.lib(Liberty) files provide standard cell characterization data with timing, power, and area model for synthesis and STA tools.

- **Timing Arcs:** Delay lookup tables vs input slew/output load
- **Sequential Timing:** Setup/Hold times for flip-flop and latches
- **Power Models:** Leakage current and dynamic switching power



Inside a .lib File

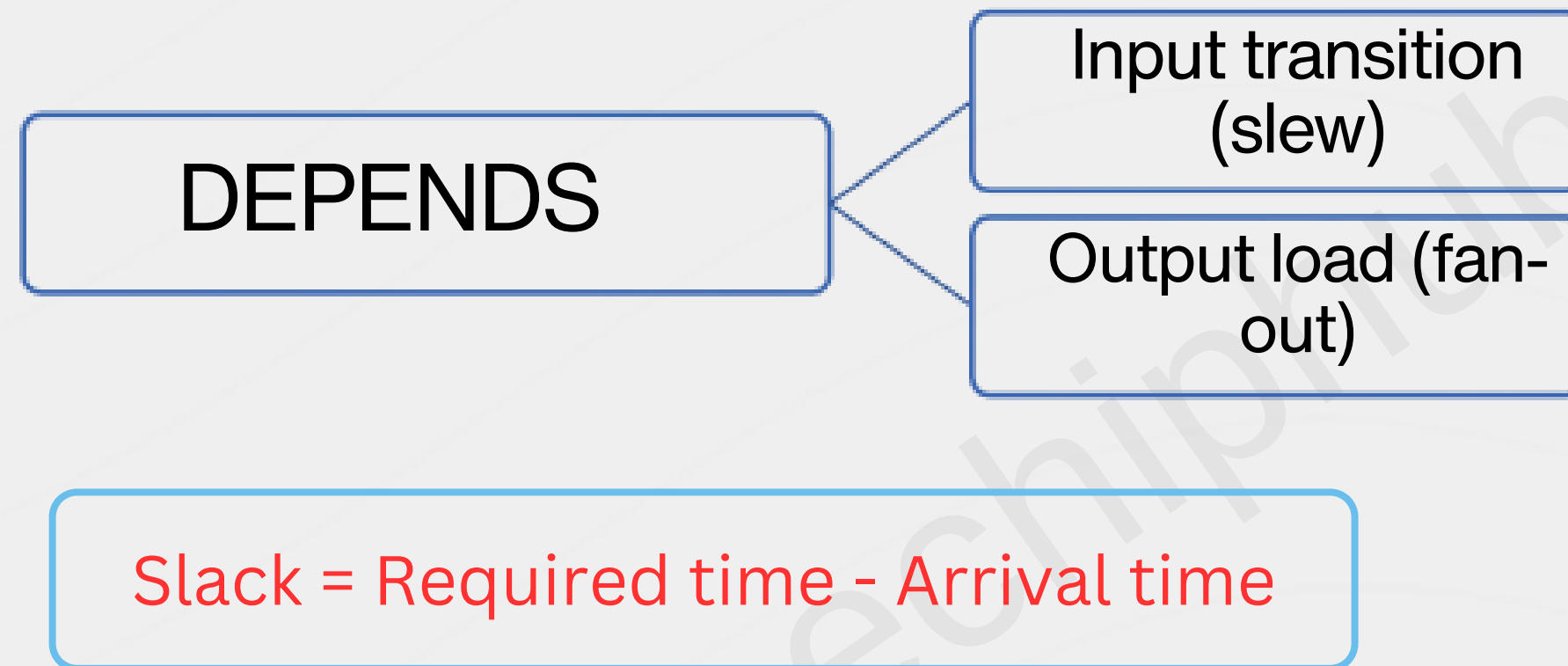
Each cell has:

- Logical function
- Input-to-output delays (timing arcs)
- Setup/hold constraints
- Slew/load dependent delay tables

Example:

```
cell (NAND2_X1) {  
  pin A { direction: input; }  
  pin Y { direction: output; timing { ... } }  
}
```

➔ Delay and Slack Calculation Using .lib



.lib provides:

- Gate delay info
- Setup/hold timing for each flip-flop

➔ SDC + .lib → Timing Closure

SDC provides:

What timing to
meet

.lib provides:

How long each
gate takes

Tool uses both to:

Calculate path delay

Optimize timing

Report slack

➔ Tips for Realistic SDC Writing

Always define clocks first

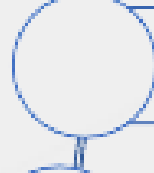
Add realistic I/O delays

Use false paths sparingly

Keep SDC portable (use get_ports, get_cells)

Review tool timing reports for missing
constraints

Summary and Best Practices

-  SDC defines timing goals using Tcl syntax
-  .lib defines cell timing and power
-  Both are essential for synthesis and STA
-  Slack is computed from arrival vs required time
-  Clean, accurate SDC = better performance + fewer surprises

These files bridge RTL and hardware reality!

Thank You

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